Software Observatory

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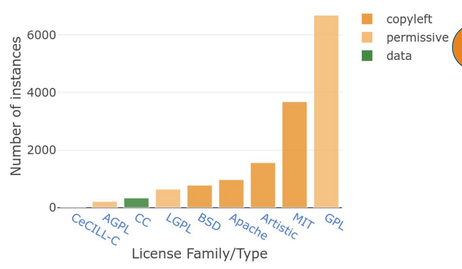
The Software Observatory aims to be an instrument for the **systematic observation** and diagnosis of the **quality of research software** in the Life Sciences. The ultimate goal is to promote the adoption of software development best practices in the Life Sciences, to which the Software Observatory can contribute through the **identification of trends** in the way research software is being developed. Indeed these trends can help **detect needs** and strategies to adopt at the **individual and community** levels.

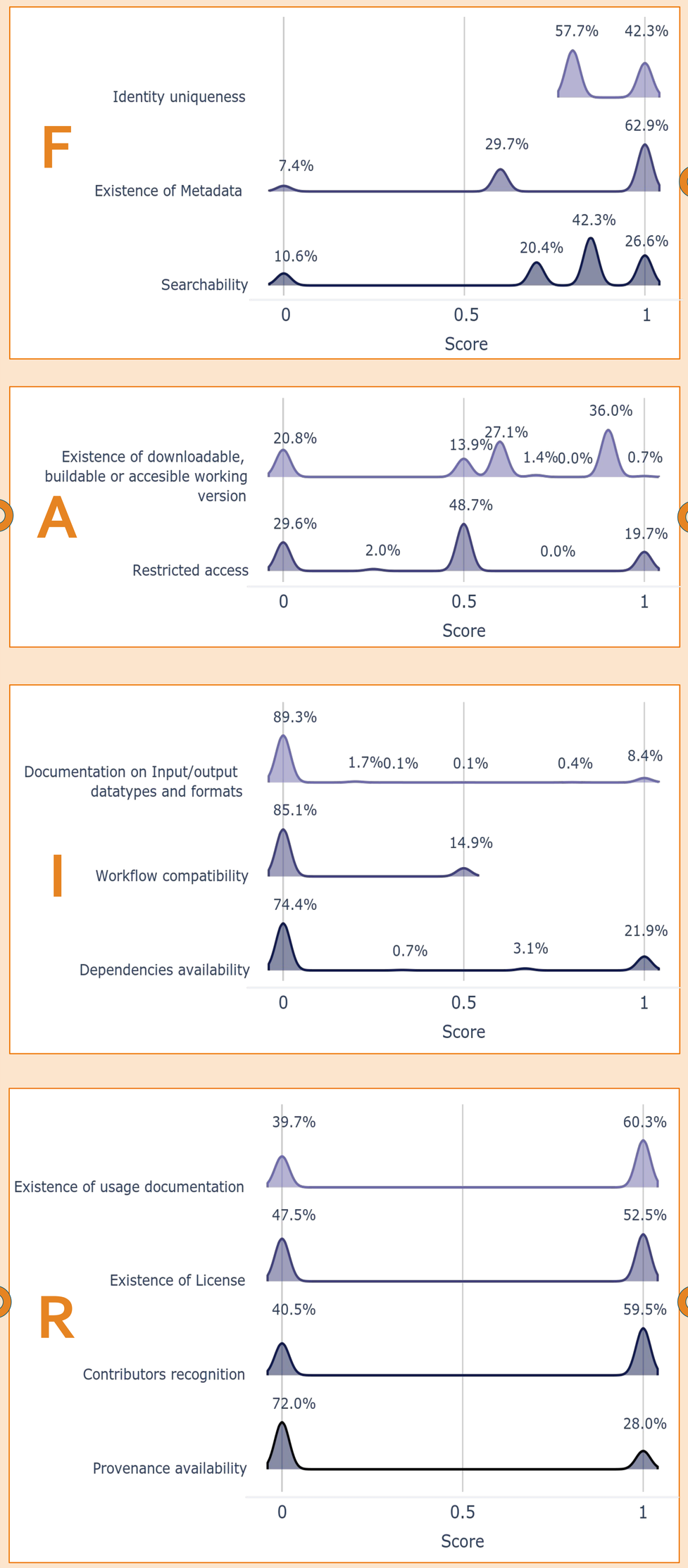
Licensing

Licensing is one of the most crucial features of a piece of software, determining both its **Accessibility** and **Reusability**.



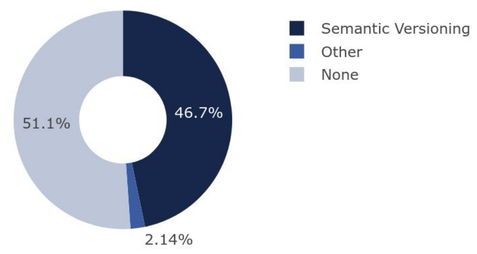
Distribution of main Open Source License families

Measurement of Research Software Quality



Versioning

Following a specific set of rules for naming software releases allows its inequivocal identification, which increases its **Findability**, **Reusability** and guarantees the **Reproducibility** of research results.

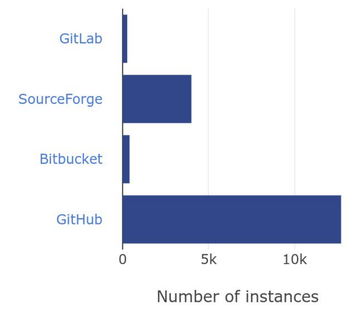


Specification of Semantic Versioning here

Repositories and Version Control

Making software available through any of the main software repositories makes it more **Findable** and **Accessible**.
Version control offers a standardized record of source code changes, making it easier to be **Reused.**

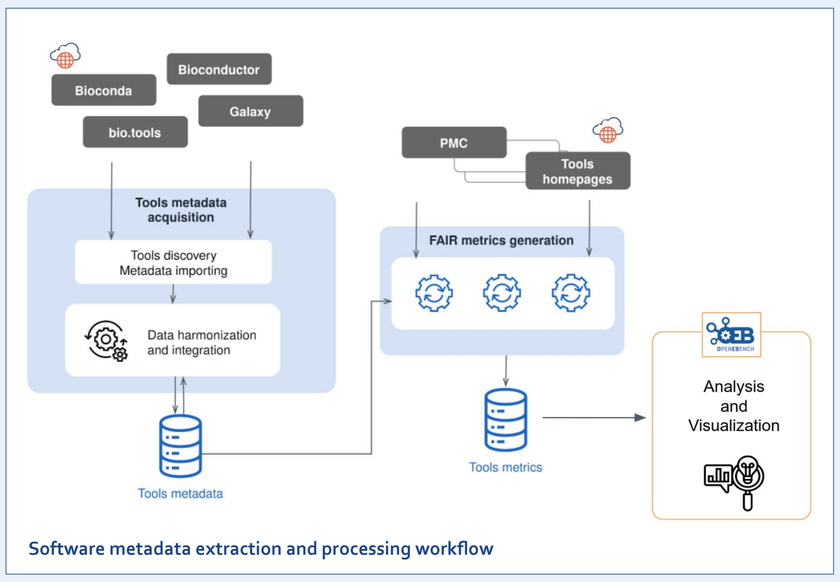
28% of instances use **version control**



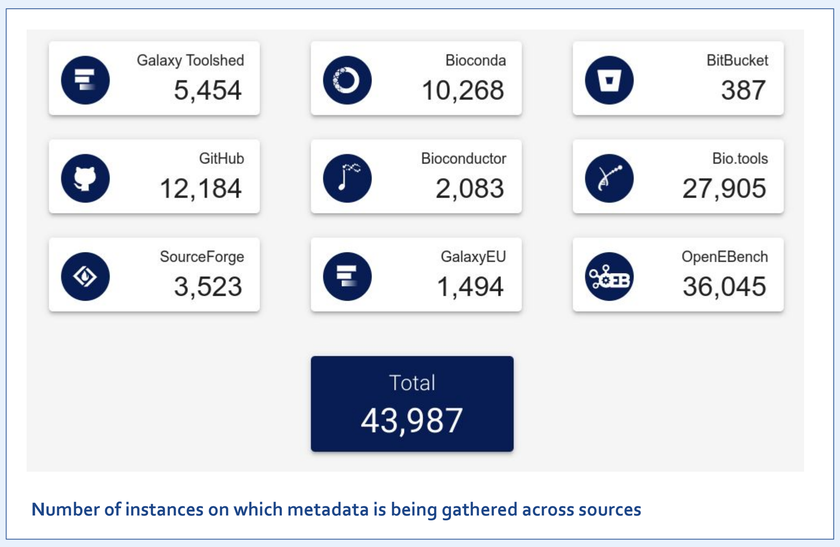
Number of instances in four widely used platforms for software distribution and development

Retrieval, transformation and integration of metadata

Automated evaluation of indicators for large amounts of software entries and the subsequent assessment of its quality requires the **retrieval, transformation and integration** of software data and metadata.



Software metadata extraction and processing workflow



Number of instances on which metadata is being gathered across sources

Conclusions

- Having better metadata describing research software in Life Sciences should facilitate the revision and development of FAIRness indicators.
- The first generation of indicators should serve to improve automated measurement algorithms, including its associated weighting.
- Periodical assessment of research software FAIRness would allow researchers to understand existing and emerging trends as well as identify areas of improvement for higher quality research software.

openebench.bsc.es/observatory

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